CSS

Grid

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Grid</title>
    <link rel="stylesheet" href="./gridPosition.css" />
  </head>
  <body>
    <!-- <div class="container">

        <div class="item div1">div 1</div>
        <div class="item div2">div 2</div>
        <div class="item div3">

            div 3
        </div>
        <div class="item div4">div 4</div>
        <div class="item div5">div 5</div>

    </div> -->

    <div class="container">
      <div class="div1">div 1</div>
      <div class="div2">div 2</div>
      <div class="div3">div 3</div>
      <div class="div4">div 4</div>
      <div class="div5">div 5</div>
      <div class="div6">div 6</div>
      <div class="div7">div 7</div>
      <div class="div8">div 8</div>
    </div>
  </body>
</html>
```

```css
/* .container > .item {
  border: 2px solid black;
  width: 100%;
  height: 100%;
  display: flex;
  justify-content: center;
```

```
  align-items: center;
}

.container {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  grid-template-rows: 150px 150px 150px;
  row-gap: 30px;
    column-gap: 30px;
  gap: 20px;
  align-items: center;
  justify-items: center;
  place-items: center;
}

.div5{
   grid-column-start: 2;
   grid-column-end: 4;
   grid-row: 2 / 4;
}

.div4{
    grid-row-start: 2;
    grid-row-end: 4;
}
 */
.container > div {
  width: 100%;
  height: 100%;
  background-color: #333;
  color: white;
  display: flex;
  justify-content: center;
  align-items: center;
}
/* .container {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  grid-template-rows: repeat(3, 100px);
  gap: 20px;
} */

/* .div1 {
  grid-row: 1 /4;
} */

/* .div2 { */
/* background-color: green; */
```

```
/* grid-column: 2/4;
}
.div6 {
  grid-column: 4;
  grid-row-start: 2;
  grid-row-end: 4;
} */

.container {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  grid-template-rows: repeat(3, 100px);
  grid-template-areas:
    "first div2 div2 div3"
    "first div4 div5 div6"
    "first div7 div8 div6";
  gap: 20px;
}
.div1 {
  grid-area: first;
}
.div2 {
  grid-area: div2;
}

.div3 {
  grid-area: div3;
}

.div4 {
  grid-area: div4;
}

.div5 {
  grid-area: div5;
}

.div6 {
  grid-area: div6;
}

.div7 {
  grid-area: div7;
}
.div8 {
  grid-area: div8;
}
```

Theory Notes

1. CSS Grid Basics

- **Grid Layout**: A two-dimensional system for arranging elements in rows and columns.
- **Key Properties**:
  - display: grid; → enables grid layout.
  - grid-template-columns → defines column structure (e.g., repeat(4, 1fr)).
  - grid-template-rows → defines row heights.
  - gap → spacing between rows and columns.

2. Positioning in Grid

- **Grid Lines**: Invisible lines that define the start and end of rows/columns.
- **Manual Placement**:
  - grid-column-start / grid-column-end → control horizontal span.
  - grid-row-start / grid-row-end → control vertical span.
  - Shorthand: grid-column: 2 / 4; or grid-row: 1 / 3;.

3. Grid Template Areas

- **Named Areas**: You can assign names to grid cells and place items by name.
- Syntax:

```
css
grid-template-areas:
  "header header header"
  "sidebar main main"
  "footer footer footer";
```

- Then assign:

```
css
.header { grid-area: header; }
.sidebar { grid-area: sidebar; }
```

4. Alignment

- justify-items → horizontal alignment of items inside cells.
- align-items → vertical alignment of items inside cells.

- place-items → shorthand for both.

5. Differences

- **Manual placement** (grid-column, grid-row) → precise control using line numbers.
- **Template areas** → semantic, easier to visualize layouts.

Practice Tasks

Coding Exercises

1. Create a grid with 3 rows and 3 columns, place one item spanning the entire first row.
2. Use grid-template-areas to design a layout with header, sidebar, main, and footer.
3. Place one item in the second column spanning rows 2 and 3.
4. Add gaps of 20px between rows and 10px between columns.
5. Experiment with place-items: center to center all items in their cells.